Faculty of Information Technology & Artificial Intelligence
Network Administration (10686213)
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Using ShellGPT as an AI Assistant for Network Administration

1. Introduction

In this task, I used **ShellGP**T with the **Ollama** local AI model on Kali Linux. The goal was to give the AI simple instructions in English and let it generate Linux commands to perform different tasks.

2. Setting Up ShellGPT

First, I installed Ollama and downloaded the Llama 3.2 model. Then, I installed ShellGPT in a Python virtual environment and configured it to use the local Ollama model.
After that, I tested ShellGPT to make sure it was working correctly. I used the **sgpt --shell** command followed by a simple request in English. ShellGPT generated a Linux command and asked me whether I wanted to execute it.

3. Testing Network Interfaces

I asked ShellGPT to show the network interfaces on my Kali Linux machine. It generated the command ip link show.
After I approved the command, it displayed the available network interfaces, including lo and eth0. This helped me check the network interfaces on my system**.**

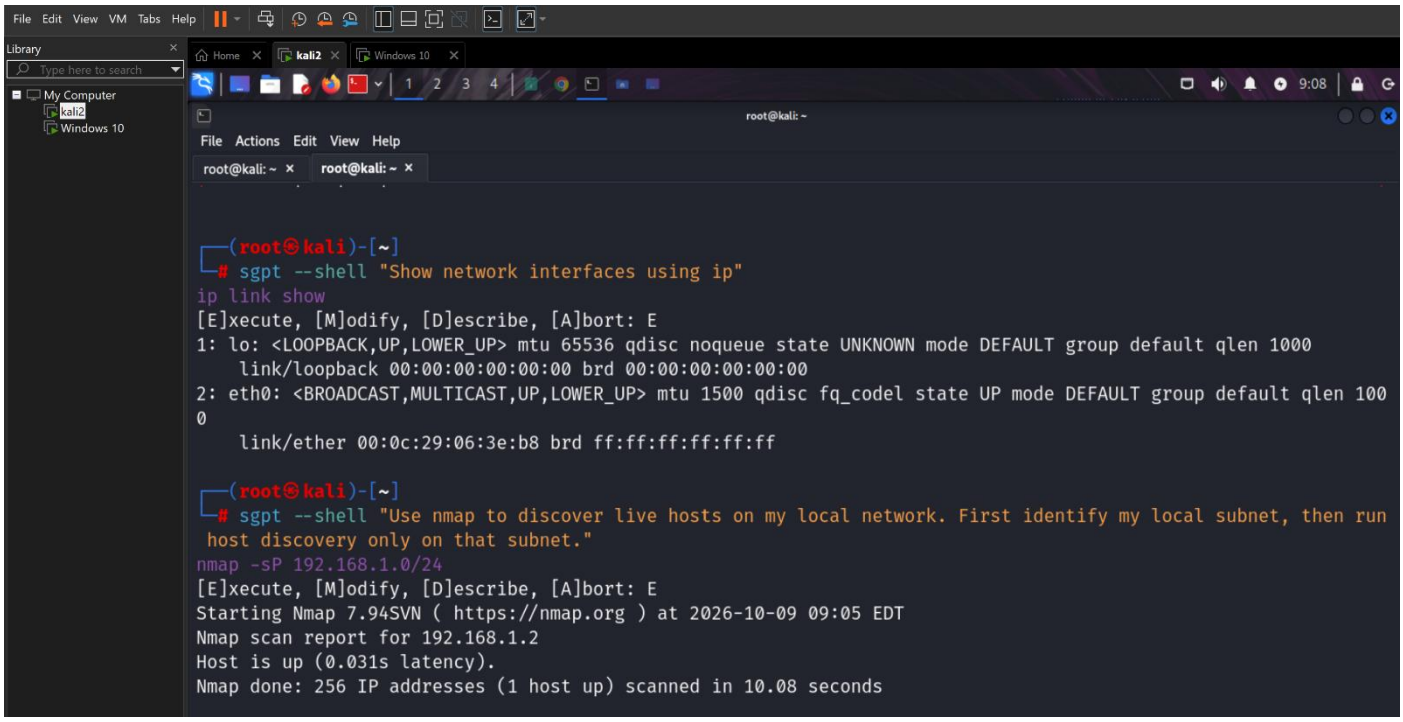
4. Checking Listening Network Ports

Next, I asked ShellGPT to check which network ports were listening on my computer. It generated the command netstat -tlnp | grep LISTEN.

After executing it, I could see the listening TCP ports and the processes using them. The output showed local services related to Ollama and the Llama server.

5. Network Host Discovery

I asked ShellGPT to use Nmap to discover active hosts on the local network. It generated a command to scan the network and identify which devices were online.





My Experience and Opinion

Personally, I would not recommend using ShellGPT for every task. It sometimes took a long time to generate commands, especially when running it inside a virtual machine. It also required a lot of system resources to run ShellGPT and the Llama 3 model together. The generated commands were not always correct, and sometimes I had to make my requests more specific to get better results. For simple tasks, it was useful, but for complex ones, checking and fixing the commands sometimes took more time than doing the task myself. Overall, it was an interesting experience, but I found it less useful for complex tasks.
